UNIVERSITY OF WAH(UOW)

DEPARTMENT OF COMPUTER SCIENCE

Lab Task # 04

COURSE: Data Structure & Algorithm Lab_201L

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<u>**Task # 01**</u>

```
int main()
{
    int arr[10], size, pos;
    cout<<"Enter Size: ";
    cin>>size;
    for(int i = 0; i < size; i++)
    {
       cin>>arr[i];
    }
    cout<<"Enter Position where you want to store: ";
    cin>>pos;
    for (int i = size; i > pos; i--)
    {
        arr[i] = arr[i-1];
    }
    size++;
    arr[pos] = 25;
    for (int i = 0; i < size; i++)
    {
        cout<<arr[i]<<" ";
    }
    return 0;
}
```

<u>**Output**</u>

```
Enter Size: 5
1 2 3 4 5
Enter Position where you want to store: 3
1 2 3 25 4 5
```

Task # 02

```
srand(time(0));
int arr[100], s, k=0;

for (int i = 0; i < 100; i++)
{
    arr[i] = rand() % 100;
    cout<<"array["<<i<<"] = "<<arr[i]<<endl;
}

cout<<"Enter the Value to search in array: ";
cin>>s;

for (int i = 0; i < 100; i++)
{
    if (s == arr[i]){
        cout<<"Number Founded: "<<arr[i]<<" at: "<<i<<endl;
        k = 1;
    }
}

if ( k == 0 ){
    cout<<"Number don't found."<<endl;
}
```

Output

```
array[89] = 1
array[90] = 13
array[91] = 1
array[92] = 62
array[93] = 58
array[94] = 42
array[95] = 31
array[96] = 53
array[97] = 18
array[98] = 38
array[99] = 98
Enter the Value to search in array: 1
Number Founded: 1 at: 74
Number Founded: 1 at: 89
Number Founded: 1 at: 91
```

Task # 03

```
srand(time(0));
int arr[20];
int max , min;
for (int i = 0; i < 20; i++)
{
    arr[i] = rand()%100;
    cout<<"Arr["<<i<<"] = "<<arr[i]<<endl;
}
 max = arr[0];
 min = arr[0];
for (int i = 0; i < 20; i++)
{
    if (max < arr[i]){
        max = arr[i];
    }
    if (min > arr[i]){
        min = arr[i];
    }
}
cout<<"Max is: "<<max<<endl;
cout<<"Min is: "<<min<<endl;
```

Output

```
Arr[10] = 86
Arr[11] = 35
Arr[12] = 16
Arr[13] = 3
Arr[14] = 98
Arr[15] = 40
Arr[16] = 10
Arr[17] = 71
Arr[18] = 50
Arr[19] = 48
Max is: 98
Min is: 3
```

Task # 04

```
int arr[10], size, pos;
cout<<"Enter Size: ";
cin>>size;
for(int i = 0; i < size; i++)
{
    cin>>arr[i];
}
cout<<"Enter Position where you want to Delete: ";
cin>>pos;
for (int i = pos; i < size; i++)
{
    arr[i] = arr[i+1];
}
size--;
for (int i = 0; i < size; i++)
{
    cout<<arr[i]<<" ";
}
return 0;
```

Output

```
Enter Size: 5
1 2 3 4 5
Enter Position where you want to Delete: 2
1 2 4 5
```

Task # 05

```
int main() {
    int n;
    cout << "Enter number of students: ";
    cin >> n;
    int marks[100];
    cout << "\nEnter marks of " << n << " students:\n";
    for (int i = 0; i < n; i++) {
        cout << "Mark " << i + 1 << ": ";
        cin >> marks[i];
    }
    int searchMark;
    cout << "\nEnter mark to search: ";
    cin >> searchMark;
```

```cpp
bool found = false;
for (int i = 0; i < n; i++) {
    if (marks[i] == searchMark) {
        cout << "Mark " << searchMark << " found at position " << i + 1 << ".\n";
        found = true;
        break;
    }
}
if (!found)
    cout << "Mark " << searchMark << " not found.\n";

int delMark;
cout << "\nEnter mark to delete: ";
cin >> delMark;

found = false;
for (int i = 0; i < n; i++) {
    if (marks[i] == delMark) {
        found = true;

        for (int j = i; j < n - 1; j++) {
            marks[j] = marks[j + 1];
        }
        n--;
        cout << "Mark " << delMark << " deleted successfully.\n";
        break;
    }
}
```

```cpp
if (!found)
    cout << "Mark " << delMark << " not found. Cannot delete.\n";

cout << "\n--- Array Information ---"<<endl;
cout << "Total memory used by array: " << sizeof(marks) << " bytes\n";
cout << "Number of elements currently stored: " << n << "\n";

cout << "\nFinal list of marks: " <<endl;
for (int i = 0; i < n; i++) {
    cout << marks[i] << " ";
}
cout << endl;

return 0;
```

Output

```
Enter marks of 5 students:
Mark 1: 22
Mark 2: 33
Mark 3: 44
Mark 4: 55
Mark 5: 66

Enter mark to search: 75
Mark 75 not found.

Enter mark to delete: 33
Mark 33 deleted successfully.

--- Array Information ---
Total memory used by array: 400 bytes
Number of elements currently stored: 4

Final list of marks:
22 44 55 66
```

Task # 07

```
int main() {
    int n;
    cout << "Enter the number of products in inventory: ";
    cin >> n;
    int quantities[100];
    cout << "\nEnter product quantities:\n";
    for (int i = 0; i < n; i++) {
        cout << "Quantity of product " << i + 1 << ": ";
        cin >> quantities[i];
    }
    int searchQty;
    cout << "\nEnter product quantity to search for: ";
    cin >> searchQty;

    bool found = false;
    for (int i = 0; i < n; i++) {
        if (quantities[i] == searchQty) {
            cout << "Quantity " << searchQty << " found at position " << i + 1 << ".\n";
            found = true;
            break;
        }
    }
    if (!found)
        cout << "Quantity " << searchQty << " not found in inventory.\n";

    for (int i = 0; i < n; i++) {
        if (quantities[i] <= 0) {
            cout << "\nProduct with quantity " << quantities[i] << " (out of stock) deleted.\n";
            for (int j = i; j < n - 1; j++) {
                quantities[j] = quantities[j + 1];
            }
```

```
    cout << "\n--- Inventory Information ---\n";
    cout << "Total memory allocated for array: " << sizeof(quantities) << " bytes\n";
    cout << "Number of products currently stored: " << n << "\n";

    cout << "\nUpdated inventory quantities:\n";
    for (int i = 0; i < n; i++) {
        cout << quantities[i] << " ";
    }
    cout << endl;

    return 0;
}
```

Output

```
Enter product quantities:
Quantity of product 1: 43
Quantity of product 2: 23
Quantity of product 3: 65
Quantity of product 4: 76
Quantity of product 5: 33
Quantity of product 6: 90

Enter product quantity to search for: 90
Quantity 90 found at position 6.

--- Inventory Information ---
Total memory allocated for array: 400 bytes
Number of products currently stored: 6

Updated inventory quantities:
43 23 65 76 33 90
```